Advanced Drug Formulation Science

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University
setu.ie

Advanced Drug Formulation Science (A25721)

Short Title:	Adv. Dru. Form. Sci.
Faculty	Science and Computing
Department	Science
Cluster	Quality, Regulatory and Compliance
Author	RJRYAN
Credits	10
Level:	Postgraduate

Description of Module / Aims

The certificate in Advanced Drug Formulation Science will provide the learner with a clear understanding of the pharmaceutical development pathway from pre-clinical to clinical and pharmaceutical development. The module will cover the physicochemical basis of pre-formulation studies and the development of dosage forms for achieving acceptable bioavailability and required efficacy. The student will gain a detailed understanding of specific drug formulation methodologies such as spray drying for poorly soluble drugs, aerosol drug formulation and oral liquid/suspension formulation. Learners will also be exposed to the area of sterilisation requirements for parenteral drugs as well as stability involving lyophilisation for protein stabilisation.

Programmes

SCIE-0077	Certificate in Advanced Drug Formulation Science (WD_SDRUG_MA)	5 / 0 / M
SCIE-0077	MSc in Analytical Science with Quality Management (WD_SANSC_R)	1 / 0 / E
SCIE-0077	PG Dip in Analytical Science with Quality Management (WD_SANSC_G)	1 / 0 / E

Indicative Content

- Pre-formulation studies with reference to ICH Q8 Pharmaceutical Development
- Development pharmaceutics: pre-formulation studies purpose and implementation, role of excipients; drug properties and the practical execution of these studies, (ratio of materials, justification for inclusion and explanation of results) with a view to section 3.2.P.2 in the Quality section of the Common Technical
- Formulation of Immediate and modified drug release profiles --/dissolution profiles
- Bioavailability and Bioequivalence considerations for generic formulations/Biowaivers.
- Process development to pilot scale and manufacturing scale up with reference to new manufacturing guideline effective 2018
- Stability requirements with reference to ICH Q1, Q6A and 6B
- Formulation development of a range of dosage forms including oral dose, parenterals, inhalations and topical formulations with particular emphasis on the principles of achieving acceptable efficacy/ bioavailability
- Blending/wet/ dry /fluid bed granulation/solid oral dose encapsulation, compression and spray drying
- Lyophilisation/Protein Stabilisation
- Aerosol drug/nasal /pulmonary formulation
- Liquid/Suspension/Emulsions
- Sterilisation requirements for parenterals and ophthalmic products

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise the principles of pharmaceutical/biopharmaceutical pre-formulation with respect to the overall design and development of drug delivery systems.
2. Evaluate how the physicochemical properties of drug molecules and excipients influence the choice of drug dosage form and subsequent formulation and manufacturing procedures.
3. Propose a formulation design and method of manufacture taking into account the physicochemical properties of drug molecules and excipients.
4. Establish method development strategies involved in lyophilisation, spray drying and aerosol drug formulation.
5. Produce a protocol for the stability testing of a pharmaceutical product.

Learning and Teaching Methods

- Classroom, Lab and Online

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4,5
Practical	35%	2, 3
In-Class Assessment	55%	1, 4
Assignment	10%	5

Assessment Criteria

<40%: The learner will demonstrate little or no understanding of the fundamentals of the learning outcomes. Will display a lack of competence with regards to practical development strategies.

40%-59%: The learner will demonstrate limited understanding of the fundamentals of the learning outcomes. Will display some ability with regards to associated practical development strategies.

60%-69%: As well as the above, the learner will demonstrate reasonable understanding of the fundamentals of the learning outcomes. Will display reasonable competence with regards to associated practical development strategies.

70%-100%: All of the above to excellent levels. In addition, the learner will demonstrate comprehensive knowledge and understanding of all learning outcomes. Will display advanced competence and independence with regards to associated practical development strategies.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	0	36
Lab	0	12
Independent Learning	0	222

Essential Material

- Allen, L.V., N.G. Popovich and H.C. Ansel. _Ansel. Pharmaceutical Dosage forms and Drug Delivery Systems_. 9th ed. London: Lippincott, Williams & Wilkins, 2011.

Supplementary Material

- "EU Legislation ." ec.europa.eu/health/documents/eudralex
- Griffin, J.P., J. Posner and G.R. Barker. _The textbook of Pharmaceutical Medicine_. London: Wiley Backwell, 2013.
- Jithan, A. _Oral Drug Delivery Technology_. India: Pharma book syndicate , 2007.
- Rowe, R.C. _Handbook of Pharmaceutical Excipients_. 7th ed. London: Pharmaceutical Press, 2012.
- Zheng, J. _Formulation and Analytical development for low dose oral drug products_. New Jersey: John Wiley, 2009.

Requested Resources

- Room Type: Chemistry Lab